# Delayed Distributed Coarsening in an Exactly Reversible Signed-Charge-Conserving Block Cellular Automaton

*A computational companion to* Zero Without Stasis*: delayed organisation and waiting-time kinetics under an exact signed balance*

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*This is the computational companion to* Zero Without Stasis*. It reports one finite formal example, a reversible artificial world whose signed total stays exactly zero while it organises, and grades every claim by the class of evidence behind it. A later causal inference from the same programme was withdrawn when its detector failed under intervention; the correction is recorded here rather than hidden. The abstract opens in plain language.*

## Abstract

We found a simple artificial world in which random material reorganises into large, irregular material-rich and near-empty regions while the net signed charge stays exactly zero and every microscopic step can be undone. The result shows how substantial organisation can develop without breaking the global balance or discarding microscopic information; in this model, its timing depends strongly on initial density and its expression depends on the local reactions. We study a deterministic four-state block cellular automaton with a frozen 256-entry local permutation. Each update conserves signed charge $Q = N_+ - N_-$ but not occupancy $M = N_+ + N_- + N_\mathrm{b}$; the studied worlds start in the global $Q = 0$ sector. A $4096 \times 4096$ localized-patch trajectory developed irregular domains and, after 65,536 forward phases and the corresponding inverse phases, recovered all 16,777,216 cells exactly. Separate $1024 \times 1024$ homogeneous-random ensembles, with 12 seeds per density, exhibited density-sensitive domain formation with high local compositional disorder. No run at initial occupancy 0.16 crossed the frozen geometrical endpoint by 262,144 phases; at 0.18, 8/12 crossed by 524,288 phases. The Kaplan–Meier median first passage decreased from 495,616 phases at 0.18 to 65,536 at 0.26. Event-centred morphology favoured distributed change rather than a localized droplet front. In the pooled control protocol, an active sibling conserving both $Q$ and $M$ and a pure-rotation rule each gave 0/16 endpoint crossings, versus 10/16 for the selected rule; several single reaction-pair removals also suppressed crossings. The contribution is this reproducible, quantitatively controlled finite-time regime, not reversibility or signed balance alone; the data are finite-time and do not establish an equilibrium phase transition.

**Keywords:** reversible cellular automata; Margolus block automata; signed charge conservation; coarsening; first passage; classifier correction

**Plain-language significance** SIGNIFICANCE

Reversible rules do not prevent macroscopic structure, but dramatic pictures alone do not identify a mechanism. This study pairs an exact cell-for-cell reversal test with waiting-time ensembles, density sweeps, arrangement controls, a matched double-conservation sibling, and reaction-pair ablations. The central result is deliberately narrow: one fully specified reversible, signed-charge-conserving automaton produces delayed, large-scale, irregular organisation from random material, and that behaviour is strongly conditioned by density and by the reaction architecture tested. All trajectories remain exactly recoverable in principle and in the large direct audit.

# 1. Introduction

This paper is the computational companion to *Zero Without Stasis* [1], which proposes exactly this kind of experiment (its Section 14.1, *Signed Life*) and states in advance what a simulation can and cannot show for that framework. The present study stands on its own as a finite computational result; the parent paper's metaphysical claims are neither assumed nor tested here.

Reversible cellular automata and lattice gases have long supplied constructive examples of local invertibility, conservation, and emergent statistical behaviour [2–6]. Conservative lattice gases recover transport and hydrodynamic phenomena [7–10]; reversible Ising and Q2R models display relaxation, long cycles, and phase-like macroscopic behaviour [11–15]; and reversible aggregation shows that invertible local dynamics can support cluster growth [16]. More recent one-dimensional reversible automata with vacancies and two particle species provide close alphabet, reaction, and conserved-charge precedents [17, 18], while reactive lattice-gas automata provide established examples of waves and pattern formation [19]. We therefore make no claim that reversibility plus organisation, multi-species reversible dynamics, or reaction-driven patterning is new in itself.

The narrower question is what can be established for a four-state signed system in which one additive quantity is exact while total occupancy can change. The distinction matters. A rule may conserve

$$Q = N_+ - N_-, \tag{1}$$

without conserving

$$M = N_+ + N_- + N_{\mathrm{b}}. \tag{2}$$

Writing "$Q = 0$" describes the global sector of the runs, not a second local conservation law and not an empty world. This paper asks whether a frozen rule in that architecture reproducibly forms large irregular domains, how density and waiting time condition the outcome, and which nearby rule controls retain or suppress it.

Our terminology is descriptive. "Coarsening-like" refers to growth of long-wavelength structure and connected regions over the observed finite lattice and time window. "Strong-domain" refers to a fixed operational endpoint. Neither term asserts equilibrium phase separation, a critical point, a free-energy description, or a universality class [20, 21].

# 2. Model and exact audits

## 2.1 Four-state partitioned dynamics

Each square-lattice cell takes a value in $\{0, +, -, \mathrm{b}\}$: empty, positive excitation, negative excitation, or bound-neutral excitation. Alternating $2 \times 2$ Margolus partitions tile the periodic

lattice. The four input symbols of each block are packed into one of 256 block states; a 256-entry permutation maps that block state to an output state. Applying the inverse permutation in reverse phase order recovers the predecessor exactly. The frozen winner's raw unsigned-byte rule table has SHA-256

`d1fe99b5735d8e9be032347de2727dc5c52a7848df200ca26f67f28c8c1bc41c`.

All 256 local cases were exhaustively audited. For every block input, the permutation is one-to-one and preserves the block contribution to $Q$. The winner does not preserve the block contribution to $M$; bound/unbound and occupied/empty reaction channels allow occupancy to change while signed balance remains exact. The construction starts from clockwise block rotation and exchanges 12 reversible pairs of outputs, so 24 of the 256 input blocks differ from pure rotation. Figure 2 shows the alternating partition and two entries from the archived table, including an actual reaction in which a balanced positive–negative pair is added while $Q$ is unchanged.

### 2.2 Large exact reversal

The audited $4096 \times 4096$ trajectory began with a random central square spanning 45% of each lattice dimension, with the rest empty. Within the patch the target state fractions were 12% positive, 12% negative and 8% bound-neutral, with exact positive–negative balance. It was advanced for 65,536 partition phases and then evolved backwards with the exact inverse schedule. The recovered lattice matched its initial state at all 16,777,216 cells. Global $Q$ remained zero throughout; $M$ changed substantially. This is an implementation and coexistence audit, not a claim that reversibility itself is novel.

## 3. Discovery, selection, and confirmation

### 3.1 A small, explicitly selective search

The winner was search-selected, not sampled blindly. The original smoke discovery evaluated 14 unique rules (16 replicate evaluations) in a reactive-rotation construction family. Its declared heuristic rewarded intermediate bands of activity, reaction density, state and block entropy, spatial and temporal mutual information, recurrence, perturbation distance, and patch repetition, with a dynamics gate and a hard charge-preservation gate. Thus subsequent behaviour cannot be described as wholly unselected.

The discovery score did *not* use the later $4096^2$ reversal result, density first-passage curves, event morphology, arrangement controls, double-conservation sibling, reaction-pair ablations, prevalence screen, or the later classifier diagnostics. Those were frozen and run as follow-up measurements. The exact search provenance and metric weights are retained with the release.

### 3.2 Separation of evidence classes

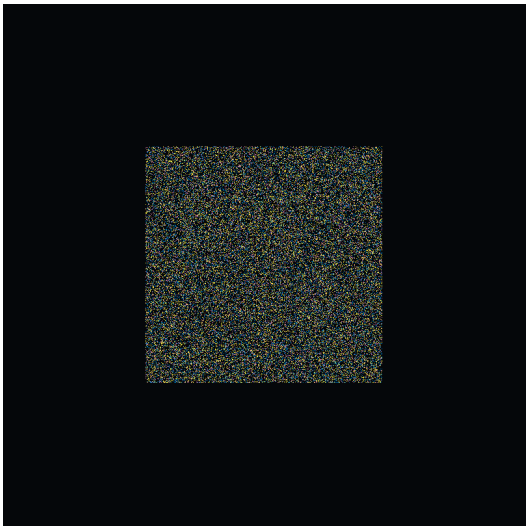
We use four labels. *Discovery* denotes the tiny search and exploratory smoke run. *Confirmation* denotes frozen winner repeats and the large exact reversal. *Mechanism constraints* denote preregistered density, geometry, sibling, and ablation comparisons. *Diagnostics* denote analyses that reveal an assay limitation and cannot support their original causal question. Results below state their class rather than pooling them as if they were exchangeable.

## Reversible dynamics, changing structure

Zero net signed charge throughout; occupancy is free to change

**A Initial localized random patch**

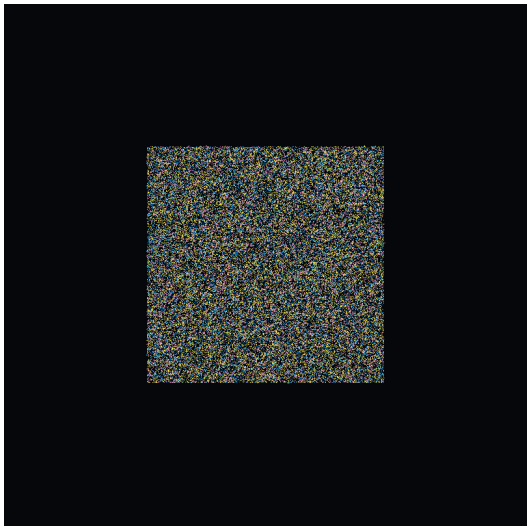
Global $Q = 0$; $M = 1{,}085{,}747$



$4096 \times 4096$ cells; phase 0

**B Same world after 20,480 phases**

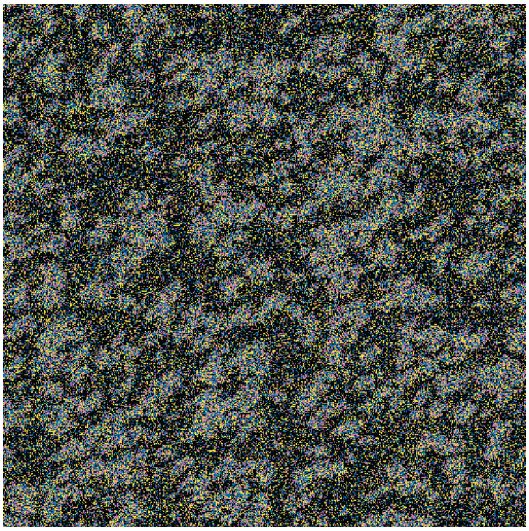
Global $Q = 0$; $M = 1{,}624{,}064$



$4096 \times 4096$ cells; phase 20,480

**C Enlarged region of B**

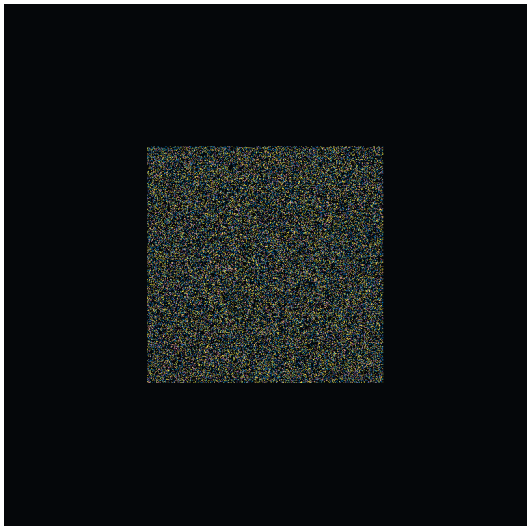
$512 \times 512$ crop at $x = 2048$, $y = 2048$



Whole-world $Q = 0$; $M = 1{,}624{,}064$

**D Initial state, recovered exactly**

Global $Q = 0$; $M = 1{,}085{,}747$



65,536 forward + 65,536 inverse phases

empty positive negative bound-neutral

**Exact reversal: zero mismatches across all 16,777,216 cells.**

**Figure 1: Structure develops while exact signed balance and reversibility are retained.** A and B show the same $4096 \times 4096$ localized-patch trajectory (seed 260824) at phases 0 and 20,480; C enlarges a $512 \times 512$ region of B. D is the archived initial-state rendering; a separate 65,536-forward/65,536-inverse-phase audit recovered that state with zero cell mismatches. The four verified source-panel PNGs are embedded directly rather than through the superseded composite-PDF rendering path. This localized-patch example is distinct from the homogeneous-start experiments.

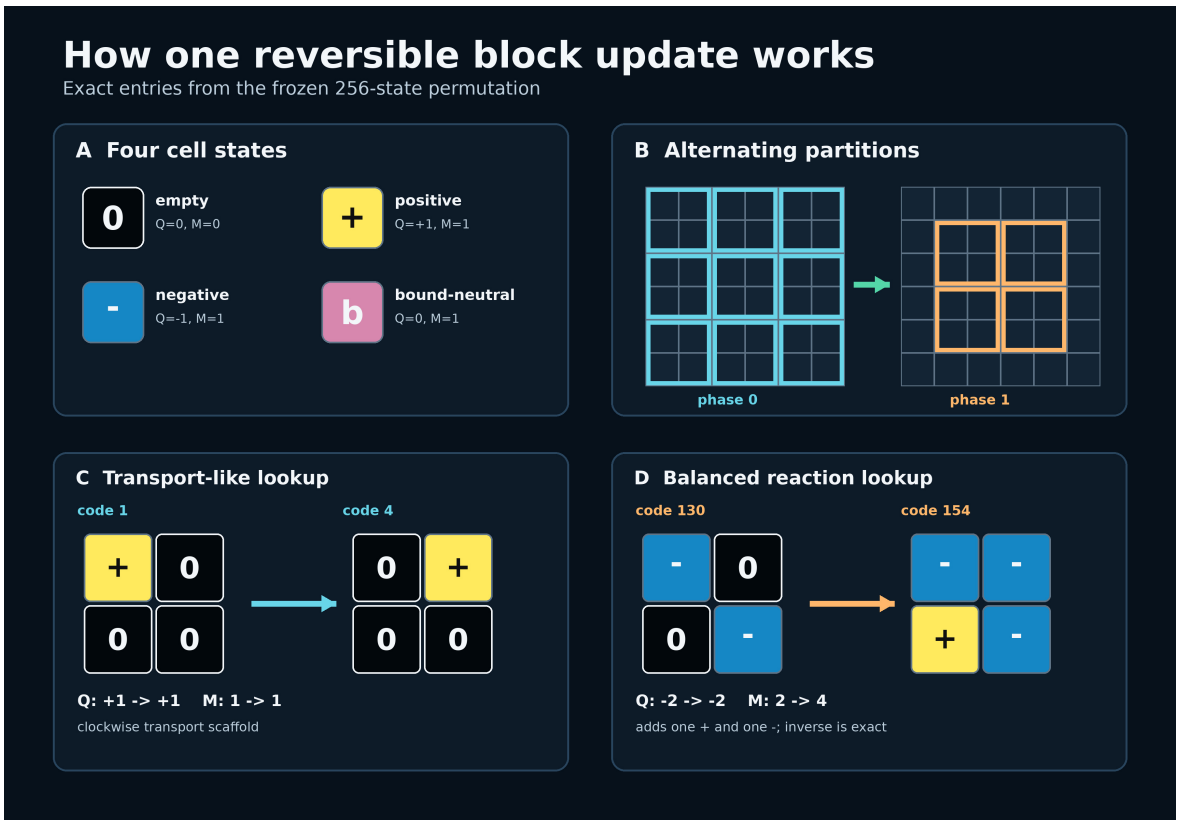



**Figure 2: How one microscopic phase works.** Panel A gives the four cell states and their contributions to signed charge $Q$ and occupancy $M$. Panel B shows the alternating $2 \times 2$ Margolus partitions. Panels C and D display entries from the actual frozen table, using cell order top-left, top-right, bottom-left, bottom-right: the transport-like entry $1 \mapsto 4$ follows clockwise block rotation, while the reactive entry $130 \mapsto 154$ changes $M$ from 2 to 4 by adding one positive and one negative excitation but preserves $Q = -2$. Because the 256-entry table is a permutation, the inverse table maps output 154 exactly back to input 130. These examples explain the mechanism without claiming that either entry alone causes the macroscopic regime. The exact codes, decoded states, invariants, and rule hash are recorded in the figure audit.

The later prevalence screen found 19 strong-domain rules among 256 independently generated charge-preserving candidates (7.4219%; Wilson 95% interval 4.8026–11.3001%). This shows that the operational regime is not unique to one table within the sampled generator, but it neither erases the original selection nor estimates prevalence in all 256! block permutations.

## 4. Measurements

The v0.3/v0.3.1 strong-domain endpoint is a geometrical, trajectory-prefix classifier; it does *not* include a spectral-growth threshold. Let $L_j$ be the measured characteristic scale of finite empty components at sampled phase $t_j$, let $r_j = L_j / \max(L_0, 1)$, and let $j^*$ be the first index attaining the largest $r_j$ in the prefix through $t_k$. Writing $f_j$ for the lattice fraction in the largest finite empty component, the implemented flag is

$$C_k = \mathbf{1}\left[r_{j^*} \geq 4, \quad r_k \geq 2, \quad f_{j^*} \geq 0.001, \quad Q_j = Q_0 \text{ for all } j \leq k\right]. \tag{3}$$

A prefix requires at least two snapshots. Spectral power, interface density, anisotropy and spatial mutual information are *separate measurements*, not additional gates in Eq. 3. This distinction is essential to the later shuffled-state false positives.

For the v0.3.1 first-passage study, $T_{\mathrm{hit}}$ is the first sampled phase with $C_k = 1$. The separate persistent endpoint, $T_{\mathrm{persistent}}$, is the first sample starting three consecutive true flags. Samples are 2,048 phases apart. A transient first hit need not be the persistent hit. Right-censored runs are retained rather than assigned artificial event times, and Kaplan–Meier medians are reported only when reached [22, 23]. Earlier campaigns used related but distinct frozen endpoints; the Supplement states their differences rather than treating all reported pass counts as interchangeable.

All quantitative claims refer to the stored sizes, seeds, phase horizons, and frozen thresholds. The endpoint is useful for comparing these runs but is not a natural law. Exact local-state images are rendered with nearest-neighbour sampling. Reported "reaction density" is the fraction of updated blocks that change reaction channel; it is not a chemical rate constant.

## 5. Results

### 5.1 Large-scale order with local compositional disorder

The localized-patch studies and the later whole-torus homogeneous-random studies both showed broad excitation-rich regions separated by near-vacuum regions, under their respective preparation and density conditions. The numerical composition result here comes from the four $4096 \times 4096$ localized-patch confirmation runs: median dense-interior occupancy at the final milestone was 0.572989, with positive, negative and bound-neutral fractions 0.192542, 0.192535 and 0.187912. Normalized four-state compositional entropy was 0.946343. High single-site diversity is not itself proof of spatial randomness; the images and separately recorded spectral diagnostics support the description of irregular, noncrystalline morphology.

Across those four localized-patch confirmatory winner runs, the transient domain-scale peak ranged from 72.66 to 136.11 times its initial value (median 98.4459), while the final recorded scale remained 15.77–16.33 times initial (median 16.1118). The effect is large but nonmonotone: the preregistered sustained late-time coarsening criterion was not met at this scale. The large transient morphology is not a success of that stronger hypothesis.

### 5.2 Density controls waiting time

The first-passage cohort comprised homogeneous-random $1024 \times 1024$ worlds. At an initial occupancy of 0.16, none of 12 runs reached the endpoint within 262,144 phases. At 0.18, 8/12 reached it in the extended 524,288-phase campaign, with median first passage 495,616 phases. All 12 replicates reached it at each tested density from 0.20 through 0.26; median first passage fell from 321,536 phases at 0.20 to 65,536 at 0.26 (Table 1). Within the v0.3/v0.3.1 endpoint, longer observation admitted events at densities that rarely crossed at shorter times. The earlier approximately 27.5% onset came from a related but different frozen classifier and is not a numerically interchangeable threshold estimate. The defensible result is strong finite-time density dependence, not an equilibrium critical density.

### 5.3 Distributed events and the prepared-geometry comparison

The frozen event-window labels classified 67 transitions as distributed, 24 as unresolved, and none as localized droplet-front events. The area comparison did not support the simple local Poisson nucleation model tested. These finite-window observations constrain a localized-front interpretation without excluding every nucleation description or establishing an asymptotic rate law.

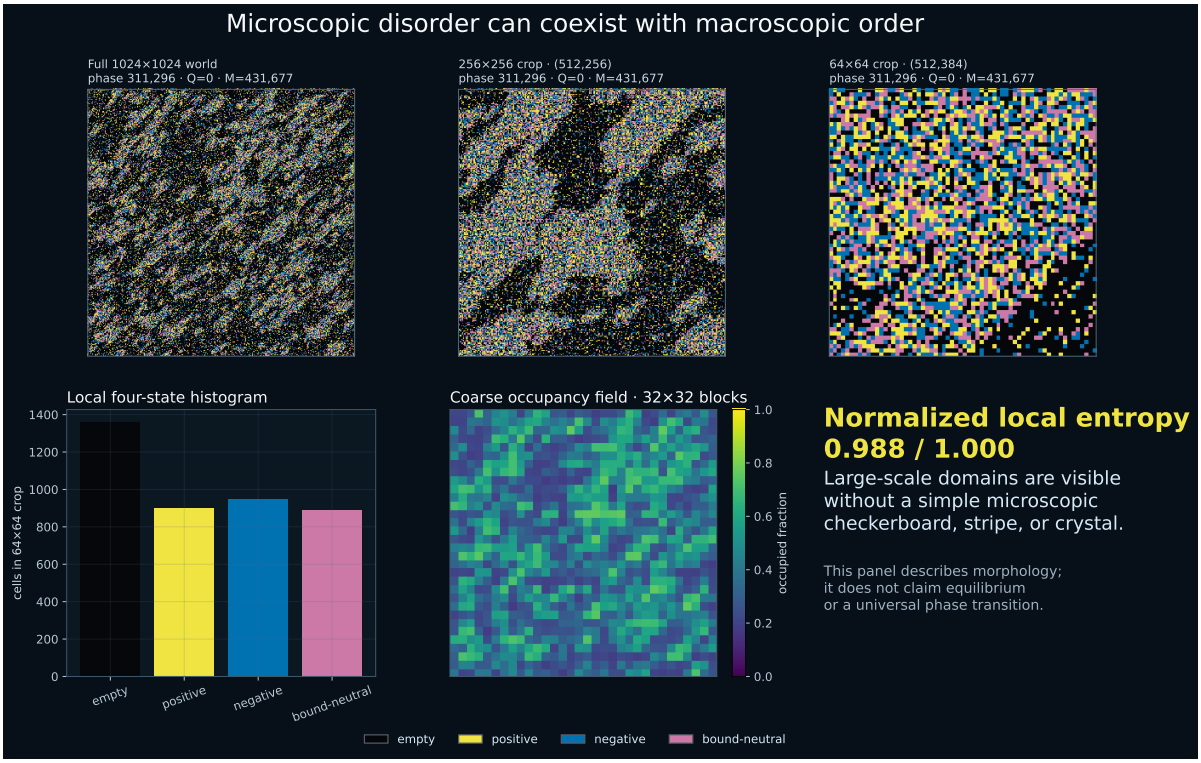



**Figure 3: Microscopic disorder and macroscopic organisation.** This representative untouched $1024^2$ homogeneous-start trajectory has initial occupancy 0.20, seed 6203 and phase 311,296; the displayed local histogram has normalized entropy 0.987741. It is distinct from the four $4096^2$ localized-patch composition averages in the text. Exact-state images and the coarse occupancy map show large-scale texture alongside a diverse local mixture. The image is a descriptive illustration, not a causal history-erasure result; high single-site entropy alone does not exclude every kind of spatial order.

**Table 1:** Homogeneous $1024^2$ first-passage cohort, with actual follow-up horizons.

| Initial occupancy | Events / 12 | KM median (phases) | Follow-up horizon (phases) |
|---:|---:|---:|---:|
| 0.16 | 0 | not reached | 262,144 |
| 0.18 | 8 | 495,616 | 524,288 |
| 0.20 | 12 | 321,536 | 262,144 / 524,288[a] |
| 0.22 | 12 | 206,848 | 262,144 |
| 0.24 | 12 | 108,544 | 262,144 |
| 0.25 | 12 | 83,968 | 262,144 |
| 0.26 | 12 | 65,536 | 262,144 |

Homogeneous $1024^2$ worlds; Kaplan–Meier median is the first sample at which the estimated survival is at most 1/2, not the median of observed events alone. [a]At occupancy 0.20, two already-transformed runs retained their 262,144-phase record and ten censored trajectories continued to 524,288. Four of the 0.18 trajectories remained right-censored. Recorded horizons, not an imputed common duration, are shown.

Prepared droplets and slabs produced 0/48 endpoint crossings by 262,144 phases, versus 19/24 for matched homogeneous starts, while their imposed occupied structures persisted or grew. Crucially, Eq. 3 measures finite empty-component growth relative to each trajectory's own initial finite-empty-domain scale; it is not an occupied-seed survival test. Preparing a large occupied shape changes the starting geometry and its normalization baseline relative to random soup. The pass-count contrast is not a common absolute test of whether a large structure forms or survives, and it cannot by itself show that seeded organisation fails or that compact seeds

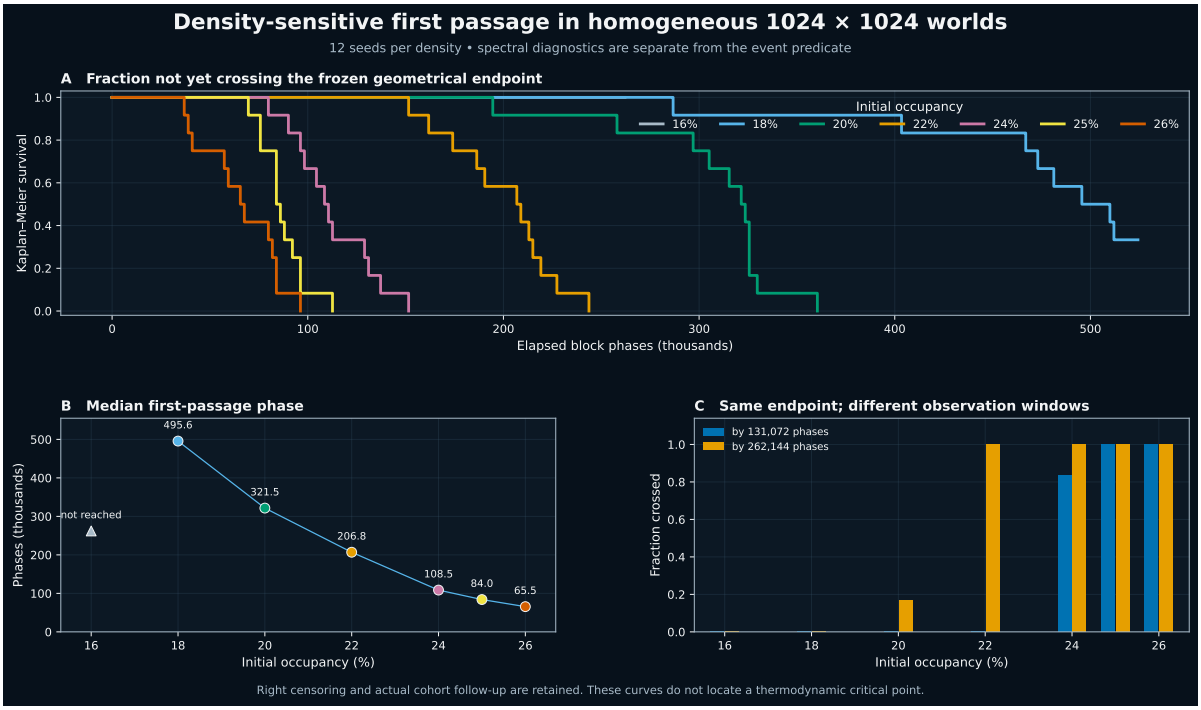



**Figure 4: Density-dependent waiting time.** Measured: right-censored survival, event-time medians, and fixed-time event fractions under the frozen endpoint. Controlled: winner rule, protocol, and replicate design across densities. Evidence class: preregistered confirmatory follow-up. Warranted inference: increasing initial occupancy substantially shortens first passage over the tested finite ensemble; longer follow-up weakens an apparent short-horizon threshold. Limitation: the curves do not locate an equilibrium critical point, establish an infinite-time threshold, or imply universal scaling.

inhibit the phenomenon. The direct persistence measurements and the relative-growth endpoint must be read separately.

### 5.4 Conservation and reaction controls constrain the mechanism

The closest reactive sibling was constructed to conserve both $Q$ and $M$ exactly while retaining nonzero reactions. It remained dynamically active (mean reaction density 0.01160 versus 0.01234 for the winner) but produced 0/8 strong-domain outcomes in the focused screen and 0/16 in the pooled balanced protocol; the winner produced 8/8 and 10/16 respectively. A pure-rotation control also produced 0/16. Thus inactivity alone does not explain the sibling result, and transport alone is insufficient under this protocol.

Single reaction-pair removals were heterogeneous. Six of twelve tested pair removals suppressed the endpoint in all 16 replicates; other removals retained 2–7 events. The architecture is therefore distributed across several channels rather than reducible, from these data, to one uniquely necessary pair. Table 2 and Figure 5 report the full pooled comparison.

## 6. Discussion

The experiments support a compact statement: a fully specified invertible block rule, restricted to the globally charge-balanced sector, can reorganise random material into enormous irregular structures after broad, density-dependent waiting times, while remaining exactly reversible cell for cell. The endpoint is accompanied by substantial long-wave growth but high local

**Table 2:** Matched controls and single-pair ablations.

| Condition | Strong-domain passes | Trials | Interpretation at tested protocol |
|---|---|---|---|
| Frozen winner (exact $Q$) | 10 | 16 | reference |
| Reactive exact-$Q{+}M$ sibling | 0 | 16 | active, no winner regime |
| Pure rotation | 0 | 16 | transport-only negative control |
| Ablations 01, 11 | 7 each | 16 each | partial retention |
| Ablation 05 | 6 | 16 | partial retention |
| Ablations 08, 09 | 3 each | 16 each | strong attenuation |
| Ablation 06 | 2 | 16 | strong attenuation |
| Ablations 00, 02, 03, 04, 07, 10 | 0 each | 16 each | suppression |

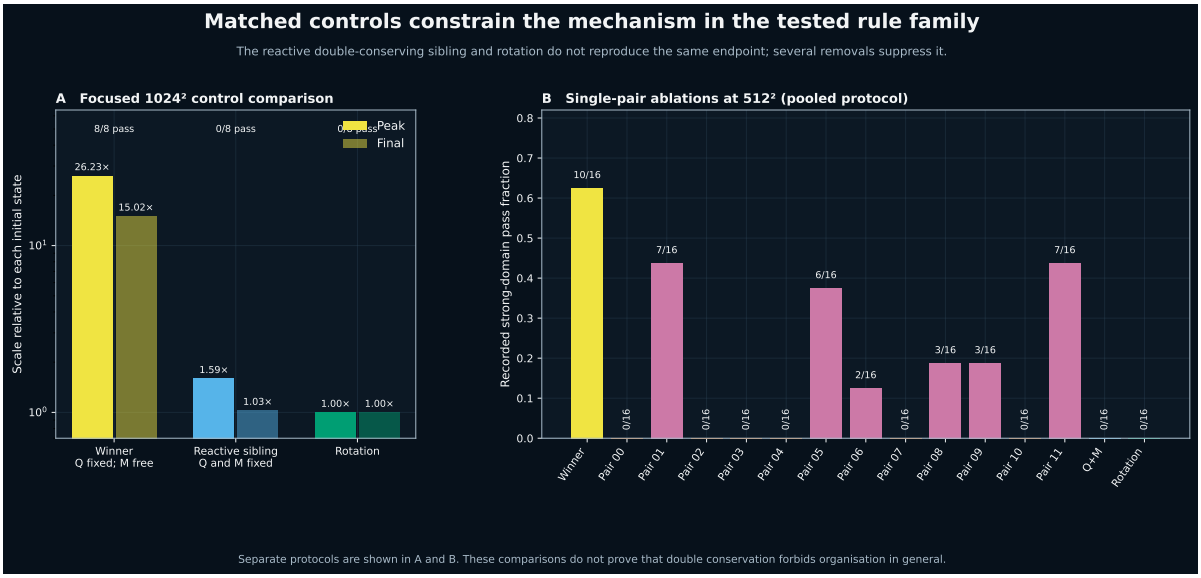



**Figure 5: Controls and ablations.** Measured: strong-domain outcomes and activity/reaction summaries for the frozen winner, exact-$Q{+}M$ sibling, rotation control, and twelve single-pair removals. Controlled: matched finite protocol and replicate counts. Evidence class: matched control and preregistered ablation. Warranted inference: the closest tested doubly conserving sibling and transport-only rule do not reproduce the winner regime, and several reaction removals strongly attenuate it. Limitation: this does not prove that variable $M$ is generally necessary, that $Q$ causes organisation, or that any one reaction is universally required.

compositional entropy. Distributed event morphology, an active exact-$Q{+}M$ sibling, transport-only controls, and reaction-pair ablations narrow the plausible mechanism without reducing it to a single reaction or established thermodynamic theory.

The closest precedents are substantive. Partitioned reversible automata supply the architecture [3, 24]; Q2R work supplies conservative reversible dynamics with macro-level relaxation [13, 14]; reversible aggregation and reversible lattice gases demonstrate aggregation and apparent macroscopic irreversibility under invertible microscopic rules [16, 25]; one-dimensional two-species reversible automata supply especially close vacancy, species, reaction, and conserved-charge precedents [17, 18]; and reactive lattice gases and reaction automata supply established pattern-forming contexts [10, 19, 26]. The residual contribution is the audited combination of a two-dimensional four-state signed invariant with variable occupancy, delayed distributed coarsening-like organisation, exact large-lattice reversal, and a frozen hierarchy of density, geometry, sibling, and reaction controls. The archived search log records query families, databases and publisher checks, and a stopping rule; it is a focused, bounded novelty review rather than proof of exhaustive priority.

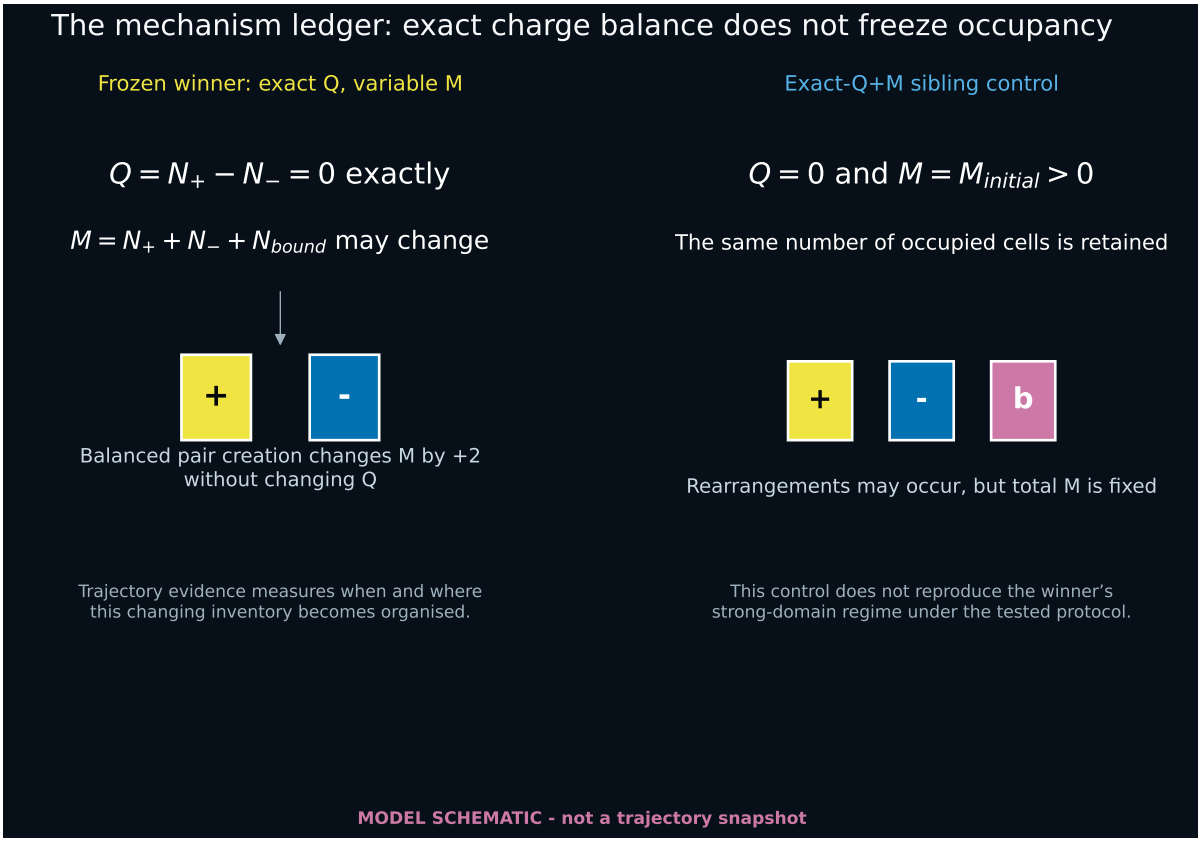



**Figure 6: Exact invariant ledger.** Measured/audited: local invariants and observed response of the winner and matched controls. Controlled: exhaustive 256-case local rule audit and explicit sibling construction. Evidence class: exact audit plus finite matched comparison. Warranted inference: $Q$ constrains one signed direction while $M$ remains an independent dynamical degree of freedom in the winner; the sibling conserves both. Limitation: the comparison does not establish a general theorem that double conservation forbids organisation.

The broad waiting-time distribution invites comparison with rare-event and first-passage analysis [22, 27]. Yet the data do not identify a free-energy barrier or stationary rate. Likewise, the separation between exact $Q$ and variable $M$ resembles, only at a conceptual level, theories coupling conserved and nonconserved fields [20, 21]. No equilibrium measure, temperature, or dynamic universality class has been demonstrated.

## 7. Limitations and withdrawn inference

**The v0.3.2 history-erasure headline** WITHDRAWN INFERENCE

The later v0.3.2 history-erasure campaign cannot support its original causal headline. Global shuffling removed more than 99% of source long-wave power, but the fixed endpoint immediately or rapidly classified many fine-grained shuffled states as "strong domain" because its connected-empty-cluster terms were confounded by random high-occupancy geometry; 9/16 primary Branch-B replicates hit at elapsed zero and 14/16 by 4,096 phases. The intervention audits and raw trajectories remain valid records, but the first-passage contrast is not a valid assay of preserved causal spatial memory. We therefore withdraw claims that the experiment showed a necessary hidden multiscale precursor or history-inhibition effect. Full diagnostics and the immutable historical report are separated in the Supplement and erratum.

More generally, the study uses one selected winner, finite two-dimensional tori, finite horizons, operational morphology thresholds, and a bounded generator prevalence sample. Three read-

ings that the vocabulary might invite are not supported by these finite-time data: that the density dependence is an equilibrium phase transition with a critical density; that the reversible yet directional-looking histories demonstrate a thermodynamic arrow; and that conserving $Q$ is what causes the organisation, rather than a constraint under which it happens to occur.

## 8. Reproducibility and evidence integrity

The release records the raw-uint8 winner hash, exact configurations, seeds, horizons, endpoint definitions, compact CSV summaries, state hashes, figure-source hashes, code version, Windows/CUDA/PyTorch environment, and an independent parity audit. For that audit the frozen rule and the Margolus update were re-implemented in Eshkol, a Scheme-based language compiled to native code [28]: the generated program re-runs the exhaustive 256-entry audit (output range, bijectivity, exact local charge) and replays a $16 \times 16$ exact-inventory torus for 256 steps, agreeing with the Python simulator cell for cell and in exact $Q$ and $M$. Large trajectories and checkpoints remain outside Git and are referenced by manifest. Every public figure is available as PDF, PNG, and SVG with SHA-256 sidecars. Historical v0.3.2 artifacts are retained byte-for-byte and accompanied by an additive erratum and corrected derivative; the repair does not rewrite the original record.

This manuscript revision is version 0.2.6. It retains the v0.2.1 methods and provenance corrections, the v0.2.2 disclosure changes, the v0.2.3 worked rule diagram and literature audit, the v0.2.4 direct embedding of the four verified source-panel PNGs of Figure 1, and the v0.2.5 archived-release DOI and house-style typesetting after *Zero Without Stasis*; it adds the citation and acknowledgement of Eshkol, the language of the independent parity oracle. The rule, detector, stored results and historical experimental records are unchanged. Figure 1 is a presentation-only rearrangement of the same archived image panels, not a new simulation or a new selection of representative states.

## 9. Code and data availability

A versioned archival deposit accompanies this paper, and a public reproducibility repository is being prepared. The release includes the frozen rule tables and controls; forward and inverse simulation code; analysis and figure-generation code; tests and environment specifications; configurations, seeds and preregistration records; compact trajectory summaries and per-run results, including failures and censored runs; the endpoint-correction record; and selected representative states with provenance and checksums. These materials should support both independent scrutiny of the reported statistics and regeneration of the experiments without requiring all full-length trajectories to be stored in Git.

**Archived code and data release:** Zenodo, DOI 10.5281/zenodo.22270456 (the companion v0.2.4 deposit; revisions v0.2.5 and v0.2.6 change presentation and metadata only).

**Public code repository:** github.com/BartlettCJ/zws-reversible-coarsening, a curated copy of the release (code, evidence tables, paper sources and figures). The archived Zenodo release remains the citable, hash-pinned source for code, data and provenance.

The release manifest identifies the exact archived version, included files and any omitted large states or trajectories. Regeneration instructions distinguish what can be reproduced directly from the compact release from what requires a new simulation run. No new scientific simulations were run for the present manuscript revision.

## 10. Conclusion

In plain terms, this artificial world builds large, irregular structures out of random material without ever upsetting its overall plus–minus balance, and its microscopic evolution can be reversed cell for cell. The structures need not be crystals: the dense regions retain high local compositional disorder.

What matters is not the arithmetic of cancellation. It is that a simple, fully specified rule produces a reproducible pattern-forming regime, with long delays that depend strongly on starting density and outcomes that change under targeted reaction controls. Fixing the net balance does not fix the amount of occupied material, its arrangement, or how those arrangements change. The active double-conservation sibling and the reaction-pair removals constrain this particular mechanism; they do not show that signed balance causes organisation or that changing occupancy is necessary in every reversible system.

The study therefore moves beyond a static example of quantities summing to zero to an experimentally characterised example of organised dynamics under that constraint.

## Acknowledgements and AI-assistance statement

Unlike the parent paper, *Zero Without Stasis*, this companion paper is the product of a mostly autonomous AI research programme, and it is described as such deliberately. Chris Bartlett supplied the motivating idea (its signed-refinement conjecture), set the objective of testing it computationally, provided the hardware, and approved the scope and the stopping rules. He did not design the experiments, write the software, run the analyses, or draft the text. Those were done by AI systems: the experimental protocols, package releases and manuscript drafts by OpenAI's ChatGPT; the simulation campaigns and repository work by OpenAI Codex agents running unattended on the author's machine; and independent critical review by Anthropic's Claude models and by OpenAI's models. The author's part was to read the stage summaries, approve or redirect each next stage, commission the reviews, decide when to stop, and deposit the release; he accepts responsibility for the claims, interpretations and any remaining errors. Supporting project and review records are retained by the author. The independent parity oracle is written in Eshkol, a Scheme-based language by tsotchke released under the MIT licence [28]; its exhaustive rule audit and cell-for-cell replay are recorded in the release.

## References

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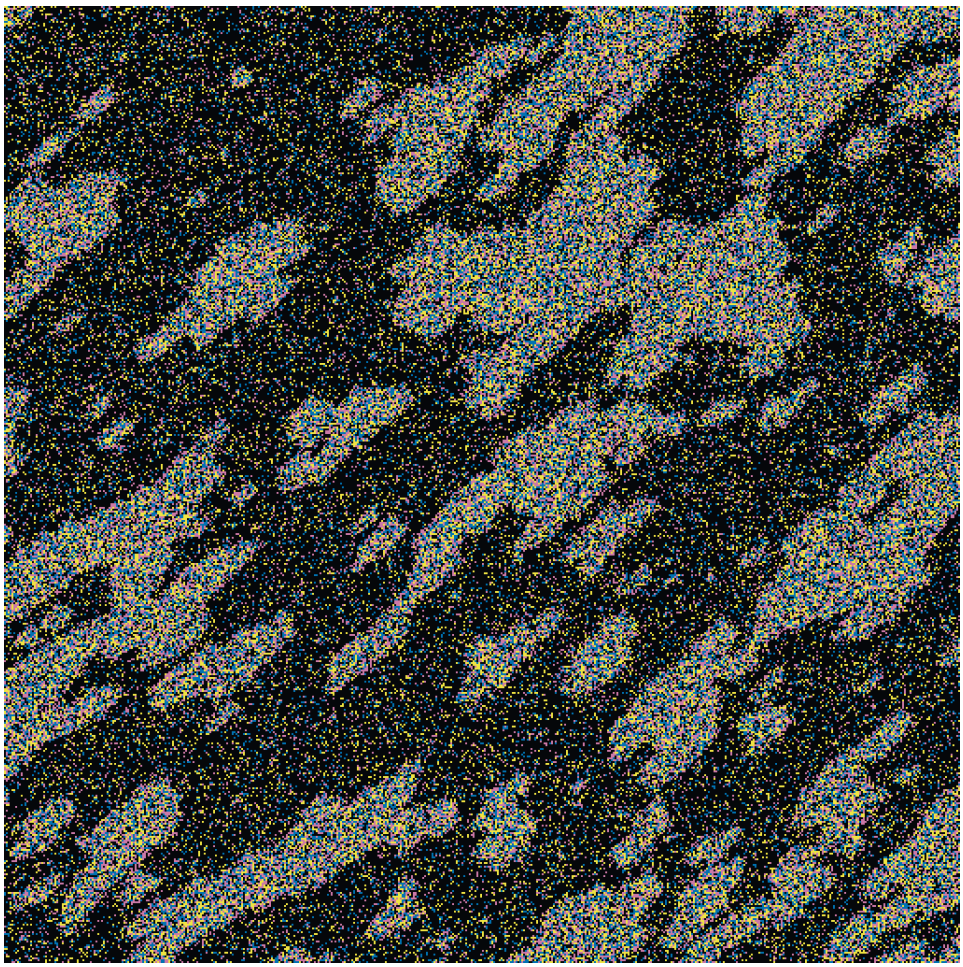
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*Zero, without stasis.*

This world began as randomly scattered material, and at every step its plus and minus cells have cancelled to exactly zero. Yet it has sorted itself into large islands of busy material and empty channels: order at the large scale, made from a mixture that is almost perfectly shuffled up close, and every step of that history could be run backwards exactly. A $512 \times 512$ window of the $1024 \times 1024$ world shown in Figure 3 (initial occupancy 0.20, seed 6203, phase 311,296), rendered cell for cell from the archived state: yellow positive, blue negative, pink bound-neutral, black empty. Global $Q = 0$; $M = 431{,}677$ of 1,048,576 cells.

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